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LOS LAB-5 Assignment

Part A: Authentication Log

Exercise 1: To print

Write sed commands to:

- Print the first log entry.
- Print the fifth log entry.
- Print entries from lines 2 to 6.
- Print the last entry.
- Print every second log entry starting from line 1.
- Print all entries containing Failed password.
- Print all successful login entries containing Accepted password.
- Print entries from the first failed login to the authentication failure.

```
anonymous@anonymous-supreme in ~/Desktop
# ) sed -n '1p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

anonymous@anonymous-supreme in ~/Desktop
# ) sed -n '5p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

anonymous@anonymous-supreme in ~/Desktop
# ) sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

anonymous@anonymous-supreme in ~/Desktop
# ) sed -n '$p' auth.log
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

anonymous@anonymous-supreme in ~/Desktop
# ) sed -n '1~2p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

anonymous@anonymous-supreme in ~/Desktop
# ) sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

anonymous@anonymous-supreme in ~/Desktop
# ) sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

anonymous@anonymous-supreme in ~/Desktop
# )
```

Exercise 2: Filtering Security Events

Write commands to:

Display failed login attempts.

Display successful login attempts.

Display entries associated with the root account.

Display entries containing Invalid user.

Display all events related to IP address 172.16.0.8.

Display all events that do not contain Accepted password.

Display all events except those generated by sudo.

Display only the entries that contain either Failed password or Invalid user using multiple -e commands.

```
anonymous@anonymous-supreme in ~/Desktop
# sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

anonymous@anonymous-supreme in ~/Desktop
# sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

anonymous@anonymous-supreme in ~/Desktop
# sed -n '/root/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

anonymous@anonymous-supreme in ~/Desktop
# sed -n '/Invalid user/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

anonymous@anonymous-supreme in ~/Desktop
# sed -n '/172.16.0.8/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

anonymous@anonymous-supreme in ~/Desktop
# sed -n '/Accepted password/!p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

anonymous@anonymous-supreme in ~/Desktop
# sed -n '/sudo/!p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

anonymous@anonymous-supreme in ~/Desktop
# sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

anonymous@anonymous-supreme in ~/Desktop
```

Exercise 3: Removing Irrelevant Entries

Without modifying the original file, display the log after:

Deleting successful login entries.

Deleting entries containing Connection closed.

Deleting the first three entries.

Deleting entries from lines 4 to 7.

Deleting all entries associated with 192.168.1.15.

Deleting every second entry starting from the first line.

Deleting all events that do not contain Failed password.

```
anonymous@anonymous-supreme in ~/Desktop
# ) sed '/Accepted password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

anonymous@anonymous-supreme in ~/Desktop
# ) sed '/Connection closed/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

anonymous@anonymous-supreme in ~/Desktop
# ) sed '1,3d' auth.log
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

anonymous@anonymous-supreme in ~/Desktop
# ) sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

anonymous@anonymous-supreme in ~/Desktop
# ) sed '/192\.168\.1\.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

```

# ) sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
anonymous@anonymous-supreme in ~/Desktop
# ) sed '/192.168.1.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
anonymous@anonymous-supreme in ~/Desktop
# ) sed '1~2d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
anonymous@anonymous-supreme in ~/Desktop
# ) sed '/Failed password/!d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

```

Exercise 4: Classifying Security Events

Replace Accepted password with [SUCCESSFUL LOGIN].

Replace Failed password with [FAILED LOGIN].

Replace Invalid user with [INVALID USER].

Replace authentication failure with [AUTHENTICATION FAILURE].

Replace root with PRIVILEGED_USER.

Replace every occurrence of server with authentication-server.

Task	Command
Accepted password → [SUCCESSFUL LOGIN]	sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log
Failed password → [FAILED LOGIN]	sed 's/Failed password/[FAILED LOGIN]/g' auth.log
Invalid user → [INVALID USER]	sed 's/Invalid user/[INVALID USER]/g' auth.log
authentication failure → [AUTHENTICATION FAILURE]	sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log
root → PRIVILEGED_USER	sed 's/root/PRIVILEGED_USER/g' auth.log
server → authentication-server	sed 's/server/authentication-server/g' auth.log

Exercise 5: Masking Sensitive Information

Use sed to anonymise the log before sharing it.

Replace IP address 10.0.0.25 with X.X.X.X.

Replace 172.16.0.8 with SUSPICIOUS_IP.

Replace username admin with USER_A.

Replace username student with USER_B.

Replace all digits with X.

Replace periods in IP addresses with hyphens.

Replace all IP addresses in the sample file with [REDACTED_IP].

Task	Command
10.0.0.25 → X.X.X.X	sed 's/10\0\0\0\25/X.X.X.X/g' auth.log
172.16.0.8 → SUSPICIOUS_IP	sed 's/172\16\0\0\8/SUSPICIOUS_IP/g' auth.log
admin → USER_A	sed 's/admin/USER_A/g' auth.log
student → USER_B	sed 's/student/USER_B/g' auth.log
All digits → X	sed 's/[0-9]/X/g' auth.log
Periods → hyphens	sed 's/\./-/g' auth.log
All IPv4 addresses → [REDACTED_IP]	sed -E 's/[0-9]{1,3}(\.[0-9]{1,3}){3}/[REDACTED_IP]/g' auth.log

Part B: Web-Server Log

Use access.log to solve the below questions

Write sed commands to:

Print all requests that returned status code 200.

Print all requests that returned status code 401.

Print all requests that returned status code 403.

Print all requests to /admin/login.

Print all POST requests.

Print requests containing ../.

Print requests containing <script>.

Print requests containing OR.

Print all requests except successful requests.

Print all requests from 10.0.0.25.

Requirement	Command
Status 200	sed -n '/ 200 /p' access.log
Status 401	sed -n '/ 401 /p' access.log
Status 403	sed -n '/ 403 /p' access.log
/admin/login	sed -n '/^admin/login/p' access.log
POST requests	sed -n '/POST/p' access.log
../ requests	sed -n '/\.\./p' access.log
<script> requests	sed -n '/<script>/p' access.log
OR requests	sed -n '/OR/p' access.log
Except successful (200)	sed -n '/ 200 /!p' access.log
From 10.0.0.25	sed -n '/^10\.\0\.\0\.\25 /p' access.log

Exercise 7: Labelling Suspicious Web Requests

Use substitution to label the events:

Replace status code 200 with [ALLOWED].

Replace status code 401 with [UNAUTHORIZED].

Replace status code 403 with [BLOCKED].

Replace /admin/login with /RESTRICTED_RESOURCE.

Replace GET with READ_REQUEST.

Replace POST with SUBMIT_REQUEST.

Original	Replacement	Command
200	[ALLOWED]	sed 's/ 200 / [ALLOWED] /g' access.log
401	[UNAUTHORIZED]	sed 's/ 401 / [UNAUTHORIZED] /g' access.log
403	[BLOCKED]	sed 's/ 403 / [BLOCKED] /g' access.log
/admin/login	/RESTRICTED_RESOURCE	sed 's/\admin\login\ /RESTRICTED_RESOURCE/g' access.log
GET	READ_REQUEST	sed 's/GET/READ_REQUEST/g' access.log
POST	SUBMIT_REQUEST	sed 's/POST/SUBMIT_REQUEST/g' access.log

Exercise 8: Grouping Security Operations

Use { } to perform multiple commands.

For entries containing status code 401:

Replace 401 with [UNAUTHORIZED].

Print the modified line.

For entries containing status code 403:

Replace 403 with [BLOCKED].

Replace the source IP address with [REDACTED_IP].

Print the modified entry.

For requests to /admin/login:

Replace the URL with /PROTECTED_PAGE.

Replace the source IP with [REDACTED_IP].

Print the result.

Requirement	Pattern	Requirement
401 → [UNAUTHORIZED] + print	/ 401 /{ s/ 401 / [UNAUTHORIZED] /; p }	401 → [UNAUTHORIZED] + print
403 → [BLOCKED] + redact IP + print	/ 403 /{ s/^[0-9.]+\/[REDACTED_IP]/; s/ 403 / [BLOCKED] /; p }	403 → [BLOCKED] + redact IP + print
/admin/login → protected + redact IP + print	/\admin\login/{ s/^[0-9.]+\/[REDACTED_IP]/; s/\admin\login\/\PROTE CTED_PAGE/; p }	/admin/login → protected + redact IP + print

Exercise 9: Adding Incident Labels

Insert the following heading before the first line of auth.log:

SECURITY AUTHENTICATION LOG

Append the following message after the final entry:

END OF SECURITY LOG

Insert [WARNING] before every failed login.

Append Review this event after every invalid-user entry.

Insert [SUSPICIOUS WEB REQUEST] before every 403 entry.

Append [REQUIRES INVESTIGATION] after every 401 entry.

Replace all entries containing Connection closed with:

[CONNECTION TERMINATED]

Replace all successful web requests with:

[NORMAL WEB REQUEST]

No.	Requirement	Command
1	Insert SECURITY AUTHENTICATION LOG before the first line	sed '1i SECURITY AUTHENTICATION LOG' auth.log
2	Append END OF SECURITY LOG after the final entry	sed '\$a END OF SECURITY LOG' auth.log
3	Insert [WARNING] before every failed login	sed '/Failed password/i [WARNING]' auth.log
4	Append Review this event after every invalid-user entry	sed '/Invalid user/a Review this event' auth.log
5	Insert [SUSPICIOUS WEB REQUEST] before every 403 entry	sed '/ 403 /i [SUSPICIOUS WEB REQUEST]' access.log
6	Append [REQUIRES INVESTIGATION] after every 401 entry	sed '/ 401 /a [REQUIRES INVESTIGATION]' access.log
7	Replace every Connection closed entry with [CONNECTION TERMINATED]	sed 's/. *Connection closed.*/[CONNECTION TERMINATED]/' auth.log
8	Replace all successful web requests (200) with [NORMAL WEB REQUEST]	sed '/ 200 /c [NORMAL WEB REQUEST]' access.log

Exercise 10: Separating Log Entries with G

Use G to:

Add a blank line after every authentication event.

Add a blank line after every failed login.

Add a blank line after every blocked web request.

Add a blank line after every second entry starting with the first entry.

No.	Requirement	Command
1	Blank line after every authentication event	<code>sed 'G' auth.log</code>
2	Blank line after every failed login	<code>sed '/Failed password/G' auth.log</code>
3	Blank line after every blocked web request	<code>sed '/ 403 /G' access.log</code>
4	Blank line after every second entry starting from first	<code>sed '1~2G' auth.log</code>

Exercise 11:

Write failed logins to failed_logins.txt.

Write successful logins to successful_logins.txt.

Write invalid-user events to invalid_users.txt.

Write 401 web events to unauthorized_requests.txt.

Write 403 web events to blocked_requests.txt.

Write all events related to 172.16.0.8 to suspicious_ip.txt.

No.	Requirement	Command
1	Failed logins → <code>failed_logins.txt</code>	<code>sed -n '/Failed password/p' auth.log > failed_logins.txt</code>
2	Successful logins → <code>successful_logins.txt</code>	<code>sed -n '/Accepted password/p' auth.log > successful_logins.txt</code>
3	Invalid users → <code>invalid_users.txt</code>	<code>sed -n '/Invalid user/p' auth.log > invalid_users.txt</code>
4	401 web events → <code>unauthorized_requests.txt</code>	<code>sed -n '/ 401 /p' access.log > unauthorized_requests.txt</code>
5	403 web events → <code>blocked_requests.txt</code>	<code>sed -n '/ 403 /p' access.log > blocked_requests.txt</code>
6	172.16.0.8 events → <code>suspicious_ip.txt</code>	<code>sed -n '/172\.16\.0\.8/p' auth.log > suspicious_ip.txt</code>

Exercise 12

Label successful and failed logins.

Label 401 and 403 web requests.

Delete successful logins and print failed logins.

Replace root with [PRIVILEGED_USER] and mask all IP addresses.

Replace GET and POST with READ_REQUEST and SUBMIT_REQUEST.

No.	Task	Command
1	Label successful and failed logins	<code>sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' -e 's/Failed password/[FAILED LOGIN]/g' auth.log</code>
2	Label 401 and 403 web requests	<code>sed -e 's/ 401 / [UNAUTHORIZED] /g' -e 's/ 403 / [BLOCKED] /g' access.log</code>
3	Delete successful, print failed logins	<code>sed -n -e '/Accepted password/d' -e '/Failed password/p' auth.log</code>
4	Replace root and mask all IPs	<code>sed -E -e 's/root/[PRIVILEGED_USER]/g' -e 's/[0-9]{1,3}(\.[0-9]{1,3}){3}/[REDACTED_IP]/g' auth.log</code>
5	Replace GET and POST	<code>sed -e 's/GET/READ_REQUEST/g' -e 's/POST/SUBMIT_REQUEST/g' access.log</code>

13 Create security_report.sed containing commands to:
Insert SECURITY INCIDENT REPORT before the first line.
Replace Failed password with [FAILED LOGIN].
Replace Invalid user with [INVALID USER ATTEMPT].
Replace Accepted password with [SUCCESSFUL LOGIN].
Replace all sample IP addresses with [REDACTED_IP].
Append END OF INCIDENT REPORT after the last line.
Execute it using:
sed -f security_report.sed auth.log
Save the output:
sed -f security_report.sed auth.log > incident_report.txt

```
anonymous@anonymous-supreme in ~/Desktop
# ) nano security_report.sed

anonymous@anonymous-supreme in ~/Desktop took 6s647ms
# ) cat security_report.sed
1i SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/g
s/Invalid user/[INVALID USER ATTEMPT]/g
s/Accepted password/[SUCCESSFUL LOGIN]/g
s/[0-9]\{1,3\}\(\.[0-9]\{1,3\}\)\{3\}/[REDACTED_IP]/g
$a END OF INCIDENT REPORT

anonymous@anonymous-supreme in ~/Desktop
# ) sed -f security_report.sed auth.log
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT

anonymous@anonymous-supreme in ~/Desktop
# ) sed -f security_report.sed auth.log > incident_report.txt

anonymous@anonymous-supreme in ~/Desktop
# ) cat incident_report.txt
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT

anonymous@anonymous-supreme in ~/Desktop
# )
```

Exercise 14: Modifying a Security Configuration File

Create security.conf:

PermitRootLogin yes

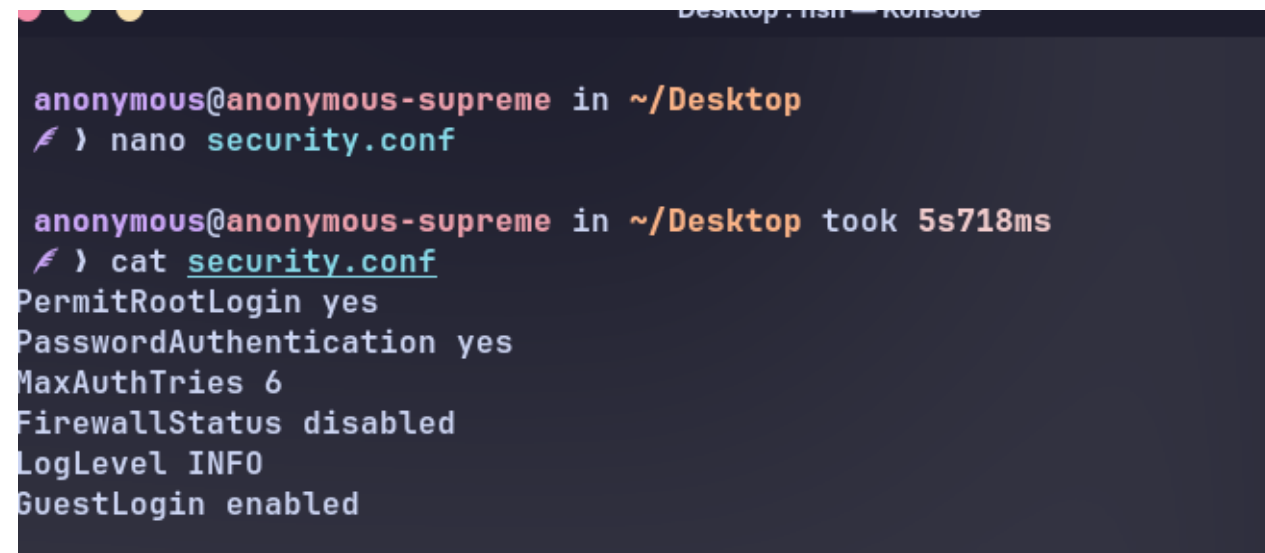
PasswordAuthentication yes

MaxAuthTries 6

FirewallStatus disabled

LogLevel INFO

GuestLogin enabled

A terminal window titled 'Desktop: fish - Konsole' shows a user named 'anonymous' at 'anonymous-supreme' in the directory '~/Desktop'. The user runs 'nano security.conf' and then 'cat security.conf'. The output of the cat command lists the configuration settings: PermitRootLogin yes, PasswordAuthentication yes, MaxAuthTries 6, FirewallStatus disabled, LogLevel INFO, and GuestLogin enabled.

```
anonymous@anonymous-supreme in ~/Desktop
✎ ) nano security.conf

anonymous@anonymous-supreme in ~/Desktop took 5s718ms
✎ ) cat security.conf
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
```

Exercise -15 First make a backup:

cp security.conf security.conf.backup

Use sed -i to:

Change PermitRootLogin yes to PermitRootLogin no.

Change MaxAuthTries 6 to MaxAuthTries 3.

Change FirewallStatus disabled to FirewallStatus enabled.

Change LogLevel INFO to LogLevel VERBOSE.

Change GuestLogin enabled to GuestLogin disabled.

```
anonymous@anonymous-supreme in ~/Desktop
# ) cp security.conf security.conf.backup

anonymous@anonymous-supreme in ~/Desktop
# ) sed -i 's/PermitRootLogin yes/PermitRootLogin no/' security.conf

anonymous@anonymous-supreme in ~/Desktop
# ) sed -i 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf

anonymous@anonymous-supreme in ~/Desktop
# ) sed -i 's/FirewallStatus disabled/FirewallStatus enabled/' security.conf
anonymous@anonymous-supreme in ~/Desktop
# ) sed -i 's/LogLevel INFO/LogLevel VERBOSE/' security.conf

anonymous@anonymous-supreme in ~/Desktop
# ) sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf

anonymous@anonymous-supreme in ~/Desktop
# ) cat security.conf
PermitRootLogin no
PasswordAuthentication yes
MaxAuthTries 3
FirewallStatus enabled
LogLevel VERBOSE
GuestLogin disabled

anonymous@anonymous-supreme in ~/Desktop
# ) cat security.conf.backup
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
```

This exercise demonstrates configuration editing only; it does not configure the actual system.

Write a Bash script named `security_log_analyser.sh`.

The script should:

Accept a log filename as a command-line argument.

Check whether the argument is provided.

Check whether the file exists.

Extract failed login attempts.

Extract invalid-user attempts.

Replace sensitive IP addresses with `[REDACTED_IP]`.

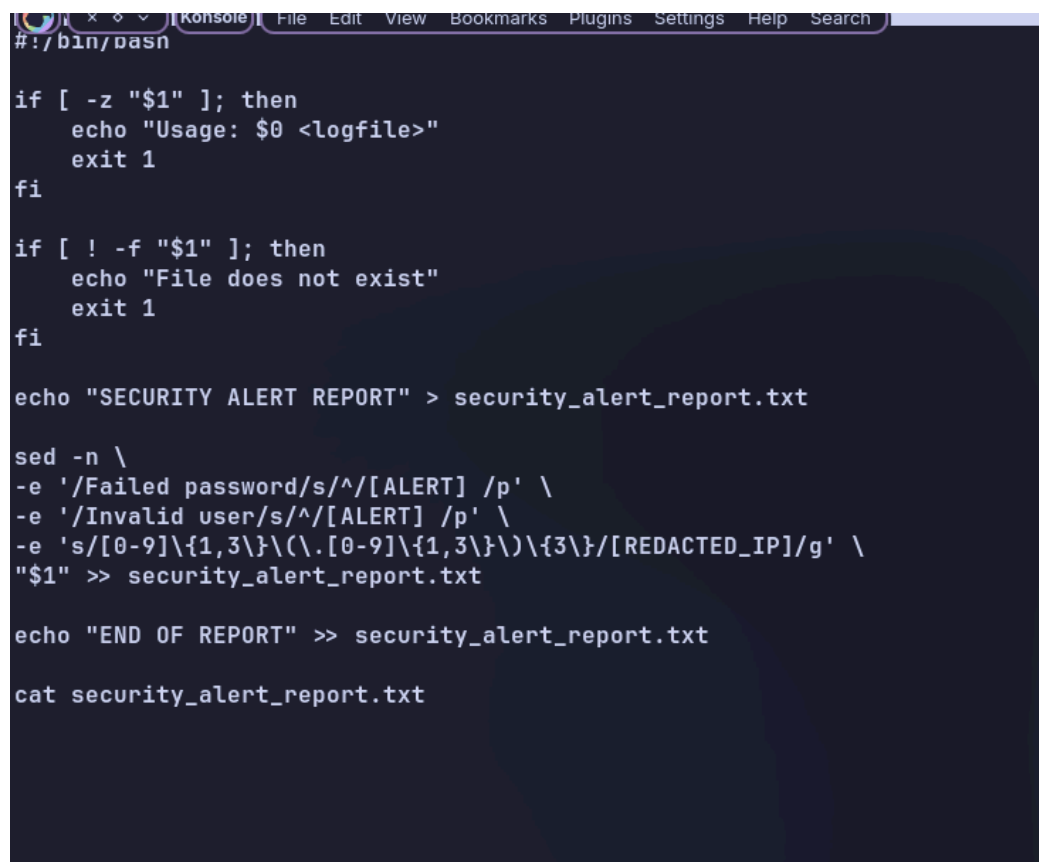
Add the heading `SECURITY ALERT REPORT`.

Add `[ALERT]` before suspicious entries.

Append `END OF REPORT`.

Save the result in `security_alert_report.txt`.

Display the generated report.



```
#!/bin/bash

if [ -z "$1" ]; then
    echo "Usage: $0 <logfile>"
    exit 1
fi

if [ ! -f "$1" ]; then
    echo "File does not exist"
    exit 1
fi

echo "SECURITY ALERT REPORT" > security_alert_report.txt

sed -n \
-e '/Failed password/s/^[ALERT] /p' \
-e '/Invalid user/s/^[ALERT] /p' \
-e 's/[0-9]\{1,3\}(\.[0-9]\{1,3\})\{3\}/[REDACTED_IP]/g' \
"$1" >> security_alert_report.txt

echo "END OF REPORT" >> security_alert_report.txt

cat security_alert_report.txt
```

```
anonymous@anonymous-supreme in ~/Desktop
[*] / nano security_log_analyser.sh

anonymous@anonymous-supreme in ~/Desktop took 3s125ms
/ ) chmod +x security_log_analyser.sh

anonymous@anonymous-supreme in ~/Desktop
/ ) ./security_log_analyser.sh
Usage: ./security_log_analyser.sh <logfile>

anonymous@anonymous-supreme in ~/Desktop
[*] / auth.log
find-the-command: "auth.log" is not found locally, searching in repositories...
find-the-command: command not found: "auth.log"

anonymous@anonymous-supreme in ~/Desktop
[Q] / cat security_alert_report.txt
File: security_alert_report.txt
SECURITY ALERT REPORT
[ALERT] Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0
.0.25
[ALERT] Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0
.0.25
[ALERT] Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.
8
[ALERT] Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172
.16.0.8
[ALERT] Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0
.0.30
END OF REPORT

anonymous@anonymous-supreme in ~/Desktop
/ ) _
```